

Richard Lara Ortiz

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Education

Princeton University Princeton, NJ
B.A. in Physics. GPA: 3.6/4.0 May 2026

- Minors in Applied and Computational Mathematics and Engineering Physics.
- Thesis title: Quantum Sensing with Superfluid Helium: Hardware Design and Phonon Transport Simulation.
- Coursework: Quantum Optics, Quantum Information & Entanglement, Optical and Quantum Devices, Quantum Computing, Photonic Systems, Machine Learning

Industry Experience

Coherent Corp. Saxonburg, PA
Electro Optic Modulator R&D Intern June to August 2026

- Designing experiments to quantify single EOM performance over a broad frequency band and find correlations for pairing two EOMs that yield reduced standing surface acoustic waves at resonant frequencies.

Research Experience

Undergraduate Senior Thesis Princeton, NJ
Advisor: Professor Stephen Lyon and Chris Tully September 2025 to May 2026

- Developed the design and layout of hardware including gold-plating, high-current printed circuit board, and cesium getters of a superfluid helium quantum sensor for the detection of low-energy neutrinos.
- Programmed a 2D Monte Carlo phonon transport simulation to model quantum evaporation efficiency in a calcium tungstate substrate coated in 30nm film of superfluid helium, including helium quasiparticle channel and interface physics selection rules.
- Implemented Monte Carlo benchmarking to test simulation, achieving 6.67% quantum evaporation efficiency, aligning with similar calculations discussed in the literature.

Physics Informed Machine Learning | Independent Work Princeton, NJ
Advisor: Professor Frederik J. Simons November 2025 to April 2026

- Applied seismic physics concepts to benchmark and improve a neural network performance of identifying p-wave arrivals, demonstrating its improved accuracy over manual methods.
- Built a peak finder algorithm to extract a subset of points starting from the predicted p-wave arrival up to a specified number of peaks and time window from a three-channel time series data.
- Engineered a Principal Component Analysis workflow to obtain polarization direction, principal components, and eigenvalues from selected time-window data to compute the rectilinearity of the p-wave pick to quantify the validity of the initial pick.

Computational Nanophotonics | Research Assistant Princeton, NJ
Advisor: Professor Alejandro Rodriguez May to August 2025

- Developed Julia real-time code dashboards to model stochastic Langevin dynamics, solve coupled-mode evolution for noisy systems, and visualize how noise affects light propagation in nonlinear optical cavities, facilitating clearer communication.
- Iteratively solved coupled stochastic differential equations and helped the group identify the optimal stochastic diffusion term that minimizes Fisher information loss in the group's inverse-design optimization workflow.

Ion-Beam Gun design for Plasma Devices | Independent Work Princeton, NJ
Advisor: Professor Christopher Tully September to December 2024

- Investigated particle accelerator design for fueling plasma devices by injecting plasma inner core with ionized gas particles.

- Utilized multiphysics software to simulate the dynamics within the accelerator by modeling charged particle transport from low to high magnetic field using a field reversed electromagnetic drift filter placed at the edge of a cyclotron magnetic field.
- Iteratively ran simulations and Python scripts to analyze and confirm stable charged-particle trajectories into high-density magnetic fields, identifying optimal charged-particle beam velocity, and feasibility of the accelerator fueling plasma devices.
- Authored a 20-page report, strengthening technical writing and physical interpretation of applied electromagnetic systems.

Nuclear Physics REU | Catholic University of America, Jefferson Lab

Washington D.C.

Advisor: Dr. Tanja Horn & Dr. Carlos Yero

May to August 2024

- Calibrated drift chamber pair in spectrometer for particle accelerator by using hydrogen-elastic collision data to address discrepancies between theoretical and experimental calculations of missing momentum at ranges higher than 700MeV/c.
- Deployed scripts in C++ with ROOT to produce overlaid normalized-histograms of theoretical and experimental measurements to facilitate data analysis and visualization in weekly group meetings.
- Delivered a 30-minute presentation to faculty and students on how my timing corrections improved the accuracy of scattered-electron measurements and inconsistencies in electron and proton dependent quantities suggesting re-optimizing optics reconstruction matrix.

Projects

Coupling Superconducting Qubits via Cavity Bus

March to May 2026

- Simulated cavity-mediated virtual photon coupling between two dispersively coupled transmon qubits using effective Hamiltonian and master-equation methods in QuTiP.
- Reproduced the AC Stark-tone-induced avoided crossing and coherent state transfer between single excitation states, showing agreement between spectroscopic splitting and time-domain oscillation frequency.

Designing Wavelength Division Multiplexing

April to May 2026

- Designed and simulated a four-channel C-band WDM Tx/Rx photonic link using silicon add-drop microring resonators aligned to 100 GHz ITU channels C47 to C44.
- Optimized microring resonance parameters with Tidy3D FDTD and SAX S-matrix models, extracting coupling coefficient, drop-port spectra, FWHM, and loaded Q-factors to evaluate channel isolation and performance.
- Laid out transmitter and receiver chips in gdsfactory/KLayout with multilayer electrical routing, common-ground buses, signal pads, via stacks, and DRC checks.

Neural Network implementation on Yelp Dataset

December 2025

- Developed a Word Embedded Long Short-Term Memory (LSTM) neural network to model the relationship between review language and star ratings, achieving a low mean average error rate of 0.449 and $R^2 = 0.759$.
- Implemented stratified 3-fold cross-validation to optimize LSTM hyperparameters (dropout rate, learning rate, and units), selecting configurations that maximized validation performance while mitigating overfitting.
- Integrated a log-odds framework on learned token representations to identify words and n-grams most highly correlated with high- and low-star reviews, enabling interpretable sentiment attribution.

Quantum Optics Experiment

December 2025

- Performed and verified the Grangier experiment, utilizing Spontaneous Parametric Down-Conversion to generate heralded photon pairs and analyzing anticorrelation parameters via coincidence counting electronics.
- Constructed and aligned a Mach-Zehnder interferometer, employing white-light spectroscopy for precise optical path-length equalization and driving a piezo-actuated mirror to modulate phase difference at the micrometer scale.
- Observed single-photon interference and executed the Quantum Eraser experiment by manipulating photon polarization state with half-wave plates to tag optical paths.

Lock-In Amplifier for Hall Resistance Measurement

October 2025

- Implemented a multi-stage analog circuit to build a lock-in amplifier circuit to measure and amplify the weak Hall voltage from a gold thin-film sensor.
- Performed oscilloscope measurements to capture and validate the amplified Hall voltage output and compute the Hall resistance with high signal fidelity.

Josephson Junction Experiment

September to October 2025

- Conducted low-temperature measurements of superconducting Josephson Junctions to analyze I-V characteristics and Shapiro step formation under microwave bias.
- Developed Python-based analysis routines (NumPy, Pandas, Matplotlib) to verify the $\pi/4$ phase relation between positive and negative voltage branches through linear regression and residual fitting.
- Performed quantitative analysis of the AC Josephson effect by modeling oscillatory residuals in the IV data, confirming frequency-voltage proportionality and demonstrating phase-coherent tunneling behavior.

Quantum Computing | Linear Combination of Unitaries

December 2024

- Derived and proved key steps of the Linear Combination of Unitaries algorithm, implementing pseudo-code and circuit representations in Qiskit to realize non-unitary operators on quantum states.
- Constructed and simulated the operator $\cos^3(\lambda Z_1 Z_2 Z_3)$ on three qubits, demonstrating oscillatory measurement probabilities and confirming theoretical predictions.

Leadership & Service

Princeton University Science Olympiad

Princeton, NJ

Boommilever Event Supervisor

October 2025 to February 2026

- Volunteering to prepare, set up, and evaluate participants' performance in a science and engineering event at the annual competition that hosts over 750+ participants from 50+ high schools.

Undergraduate Tutor, Department of Physics

Princeton, NJ

PHY301 Thermodynamics

September to December 2025

PHY208 Quantum Mechanics

February to May 2025

- Host 2+ hours of 1-on-1 tutoring each week, using problem-solving and learning heuristics to assist students' understanding.
- Review course material and guide students' logic and reasoning of solving homework problems.

The Gates Scholarship Summer Institute (TGSSI) Mentor

May to June 2025

- Volunteered to mentor a group of incoming scholars and lead an in-person component for the scholar summer institute.

Teaching Assistant, EGR150 Foundations of Engineering

June 2023 to August 2023

- Managed 20+ weekly review hours for an introductory calculus-based engineering course for incoming low-income and/or first-generation, first-year students.
- Taught basic foundations to CREO and MATLAB to help students design 3D models for their final project and gain familiarity with computational software.

Skills

Computer: Python (Qiskit, QuTiP, Matplotlib, Sklearn, NumPy, Keras, pandas, CVXPY), MATLAB, C++ (ROOT), Julia

Tools: GitHub, Jupyter Notebook, VSCode, GDSFactory, KLayout, Tidy3D, LaTeX, KiCad, AutoCAD

Languages: Spanish (Professional), Chinese (Intermediate)

Awards & Honors

Allen G. Shenstone Prize for Physics, Princeton University

2026

- Given to outstanding Physics undergraduates who have shown excellence in course work and promise in independent research.

HSF Scholar, The Hispanic Scholarship Fund

2022, 2023

- Merit-based award; selected as 1 of 10,000 from a pool of 124,000+ applicants.

The Gates Scholarship, Questbridge Scholarship

2022